

# SQL & POWER BI SALES ANALYSIS

Portfolio Project

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SQL • Power BI • Data Analysis • Data Visualization

## TOOL

DB Browser for SQLite • Power BI Desktop

## DATABASE

SQLite — sample sales database

## FOCUS

SQL querying, DAX measures and dashboard reporting

# Project Overview

A compact SQL and Power BI project demonstrating practical querying, data modeling and dashboard reporting across product, customer and order data.

**Objective.** Use SQL to explore a small sales database, answer business-style questions, and demonstrate core querying skills including filtering, sorting, joins, aggregation, grouping and calculated revenue. Then use Power BI to model the same data and communicate key sales metrics through an interactive dashboard.

## SQL skills demonstrated

SELECT • WHERE • ORDER BY • LIMIT • INNER JOIN • GROUP BY • HAVING • COUNT • SUM • AVG • calculated fields

## DATABASE STRUCTURE

| Table     | Key columns used                                                  | Role in analysis                     |
|-----------|-------------------------------------------------------------------|--------------------------------------|
| customers | customer_id, customer_name, segment                               | Customer identity and segmentation   |
| products  | product_id, product_name, category, unit_price                    | Product and pricing information      |
| orders    | order_id, customer_id, product_id, quantity, status, discount_pct | Transaction details and order status |

## RELATIONSHIPS

customers.customer\_id → orders.customer\_id

products.product\_id → orders.product\_id

# SQL Analysis

Queries 01–02

## 01 Product Price Overview

List all products with their category and price.

SQL QUERY

```
SELECT product_name, category, unit_price  
FROM products;
```

RESULT PREVIEW

|   | product_name             | category    | unit_price |
|---|--------------------------|-------------|------------|
| 1 | Wireless Mouse           | Accessories | 59.9       |
| 2 | Mechanical Keyboard      | Accessories | 189.0      |
| 3 | USB-C Hub                | Accessories | 129.0      |
| 4 | 24-inch Monitor          | Electronics | 499.0      |
| 5 | Laptop Stand             | Office      | 89.0       |
| 6 | Webcam                   | Electronics | 219.0      |
| 7 | Noise-Cancelling Headset | Electronics | 329.0      |

**Finding** The dataset contains products across Accessories, Electronics, Office and Storage categories.

## 02 Products Above RM200

Find all products priced above RM200.

SQL QUERY

```
SELECT product_name, unit_price  
FROM products  
WHERE unit_price > 200;
```

RESULT PREVIEW

|   | product_name             | unit_price |
|---|--------------------------|------------|
| 1 | 24-inch Monitor          | 499.0      |
| 2 | Webcam                   | 219.0      |
| 3 | Noise-Cancelling Headset | 329.0      |
| 4 | External SSD 1TB         | 399.0      |
| 5 | Ergonomic Chair          | 699.0      |

**Finding** Five products in the sample dataset are priced above RM200.

# SQL Analysis

Queries 03–04

## 03 Top 5 Most Expensive Products

List the five most expensive products, ordered from highest to lowest price.

SQL QUERY

```
SELECT product_name, unit_price
FROM products
ORDER BY unit_price DESC
LIMIT 5;
```

RESULT PREVIEW

|   | product_name             | unit_price |
|---|--------------------------|------------|
| 1 | Ergonomic Chair          | 699.0      |
| 2 | 24-inch Monitor          | 499.0      |
| 3 | External SSD 1TB         | 399.0      |
| 4 | Noise-Cancelling Headset | 329.0      |
| 5 | Webcam                   | 219.0      |

**Finding** The Ergonomic Chair is the highest-priced product shown at RM699.

## 04 Completed Orders with Customer Names

List all completed orders together with the customer names.

SQL QUERY

```
SELECT customer_name, status
FROM customers
INNER JOIN orders
ON customers.customer_id = orders.customer_id
WHERE status = 'Completed';
```

RESULT PREVIEW

|    |             |           |
|----|-------------|-----------|
| 41 | Mei Ling    | Completed |
| 42 | Sara Lim    | Completed |
| 43 | Amirul Faiz | Completed |
| 44 | Priya Nair  | Completed |
| 45 | Irfan Malik | Completed |
| 46 | Chloe Wong  | Completed |

```
Execution finished without errors.
Result: 46 rows returned in 17ms
At line 48:
SELECT customer_name, status
FROM customers
```

**Finding** The join returns 46 completed orders and links each order to its customer.

# SQL Analysis

Queries 05–06

## 05 Order Count by Status

Find the number of orders for each order status.

### SQL QUERY

```
SELECT status, COUNT(status)
FROM orders
GROUP BY status;
```

### RESULT PREVIEW

|   | status    | count(status) |
|---|-----------|---------------|
| 1 | Cancelled | 2             |
| 2 | Completed | 46            |

**Finding** The sample contains 46 completed orders and 2 cancelled orders.

## 06 Quantity Sold by Product

Find the total quantity sold for each product, excluding cancelled orders.

### SQL QUERY

```
SELECT product_name, SUM(quantity) AS total_quantity
FROM products
INNER JOIN orders
ON products.product_id = orders.product_id
WHERE status != 'Cancelled'
GROUP BY product_name;
```

### RESULT PREVIEW

|    |                          |    |
|----|--------------------------|----|
| 4  | External SSD 1TB         | 8  |
| 5  | Laptop Stand             | 10 |
| 6  | Mechanical Keyboard      | 9  |
| 7  | Noise-Cancelling Headset | 6  |
| 8  | USB-C Hub                | 9  |
| 9  | Webcam                   | 5  |
| 10 | Wireless Mouse           | 8  |

Execution finished without errors.  
Result: 10 rows returned in 16ms  
at line 54.

**Finding** Quantities are aggregated at product level while cancelled orders are excluded.

# SQL Analysis

Queries 07–08

## 07 Average Price by Category

Find the average product price for each category.

### SQL QUERY

```
SELECT category, AVG(unit_price) AS avgunitprice
FROM products
GROUP BY category;
```

### RESULT PREVIEW

|   | category    | avgunitprice     |
|---|-------------|------------------|
| 1 | Accessories | 125.966666666667 |
| 2 | Electronics | 349.0            |
| 3 | Office      | 289.0            |
| 4 | Storage     | 399.0            |

Execution finished without errors.

**Finding** Average prices are RM125.97 for Accessories, RM349 for Electronics, RM289 for Office and RM399 for Storage.

## 08 Completed-Order Revenue by Product

Calculate the total revenue generated by each product from completed orders, taking discounts into account.

### SQL QUERY

```
SELECT product_name,
       SUM(quantity * unit_price * (1 - discount_pct /
100.0)) AS total_revenue
FROM products
INNER JOIN orders
ON products.product_id = orders.product_id
WHERE status = 'Completed'
GROUP BY product_name;
```

### RESULT PREVIEW

|   | product_name        | total_revenue |
|---|---------------------|---------------|
| 1 | 24-inch Monitor     | 3288.41       |
| 2 | Desk Lamp           | 632.0         |
| 3 | Ergonomic Chair     | 3928.38       |
| 4 | External SSD 1TB    | 2996.49       |
| 5 | Laptop Stand        | 881.1         |
| 6 | Mechanical Keyboard | 1507.6        |

Execution finished without errors.  
Result: 10 rows returned in 21ms  
At line 61:  
SELECT product\_name,

**Finding** The calculation combines quantity, unit price and discount percentage before aggregating revenue by product.

# SQL Analysis

Queries 09–10

## 09 Customers with More Than 3 Completed Orders

Find customers who have placed more than three completed orders.

### SQL QUERY

```
SELECT customer_name, COUNT(orders.order_id) AS
completed_orders
FROM customers
INNER JOIN orders
ON customers.customer_id = orders.customer_id
WHERE status = 'Completed'
GROUP BY customers.customer_id, customer_name
HAVING COUNT(orders.order_id) > 3;
```

### RESULT PREVIEW

|    | customer_name | completed_orders |
|----|---------------|------------------|
| 5  | Nadia Ismail  | 4                |
| 6  | Mei Ling      | 4                |
| 7  | Sara Lim      | 4                |
| 8  | Amirul Faiz   | 4                |
| 9  | Priya Nair    | 4                |
| 10 | Irfan Malik   | 4                |
| 11 | Chloe Wong    | 4                |

**Finding** The result identifies repeat customers who meet the threshold of more than three completed orders.

## 10 Completed-Order Revenue by Customer Segment

Calculate the total revenue from completed orders for each customer segment.

### SQL QUERY

```
SELECT segment, SUM(quantity * unit_price) AS
total_revenue
FROM customers
INNER JOIN orders
ON customers.customer_id = orders.customer_id
INNER JOIN products
ON orders.product_id = products.product_id
WHERE status = 'Completed'
GROUP BY segment;
```

### RESULT PREVIEW

|   | segment   | total_revenue |
|---|-----------|---------------|
| 1 | Corporate | 9559.0        |
| 2 | Retail    | 9252.2        |

Execution finished without errors.  
Result: 2 rows returned in 9ms  
At line 38:

```
SELECT segment, sum(quantity * unit_price) AS total_revenue
FROM customers
```

**Finding** Completed-order revenue totals RM9,559.00 for Corporate and RM9,252.20 for Retail in the sample output.

## Key Findings & Skills

A concise summary of what the project demonstrates.

### Key findings

- 46 of 48 orders in the sample output are completed; 2 are cancelled.
- The Ergonomic Chair is the highest-priced product in the top-five query at RM699.
- Category-level average prices range from approximately RM125.97 to RM399.
- Revenue analysis uses expressions, joins and aggregation to summarize performance by product and customer segment.
- The project identifies repeat customers using GROUP BY together with HAVING.

### Capabilities demonstrated

|                                |                                                                              |
|--------------------------------|------------------------------------------------------------------------------|
| <b>Data retrieval</b>          | Selecting relevant columns and presenting concise result sets.               |
| <b>Filtering &amp; sorting</b> | Applying WHERE, ORDER BY and LIMIT to isolate and rank records.              |
| <b>Relational querying</b>     | Joining customers, orders and products through key relationships.            |
| <b>Aggregation</b>             | Using COUNT, SUM and AVG with GROUP BY and HAVING.                           |
| <b>Calculated metrics</b>      | Building revenue expressions that account for quantity, price and discounts. |

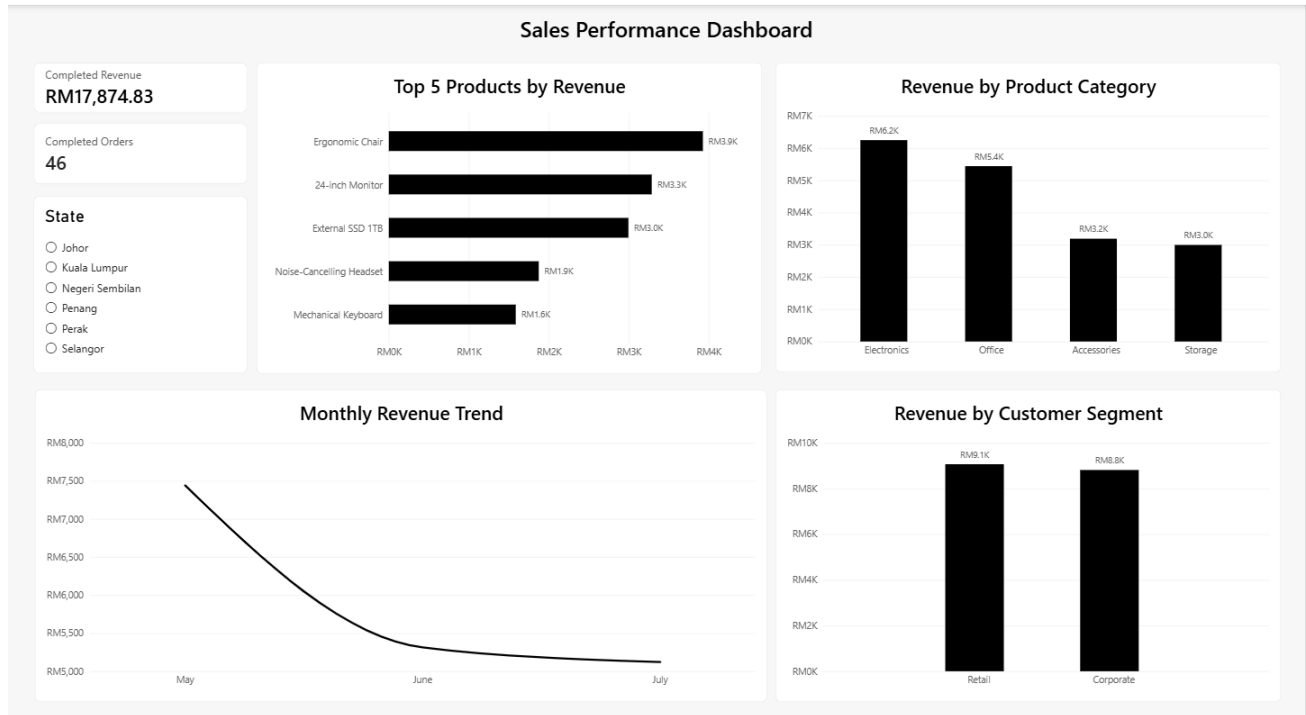
*All queries were executed in DB Browser for SQLite using the project sample database.*

# Power BI Dashboard

## Interactive Sales Performance Reporting

POWER BI • DAX • DATA MODELING • INTERACTIVE FILTERING

The dashboard uses the same customers, products and orders data analyzed in SQL, with relationships between the tables and DAX measures for completed-order metrics. It presents the main sales results in a concise, interactive view.



Sales Performance Dashboard — Power BI Desktop

## Dashboard highlights

|                          |                                                       |
|--------------------------|-------------------------------------------------------|
| <b>KPI measures</b>      | Completed Revenue and Completed Orders                |
| <b>Performance views</b> | Top 5 Products, Product Category and Customer Segment |
| <b>Trend analysis</b>    | Monthly completed-order revenue trend                 |
| <b>Interactivity</b>     | State slicer filters all report visuals               |